

Chloride, Not BPG

The quiet chemical accident that made a cow's hemoglobin a better starting point than a person's

Ask a room of physicians why a person's blood carries oxygen and most will say hemoglobin. Ask why hemoglobin outside a red cell fails at the job, and the room grows quieter. The answer is a small molecule that most clinicians met once in physiology class and forgot: 2,3-bisphosphoglycerate, usually abbreviated BPG or, in older texts, DPG.

This essay is about that molecule, about its absence from a cow's red cells, and about how that absence turned bovine hemoglobin into the raw material of choice for the oxygen carriers that reached the clinic. It is a story about allostery, the way a protein changes shape in response to something binding at a distant site, and it is one of the cleaner examples I know of nature doing a favor to an engineer without meaning to.

The problem with naked human hemoglobin

A red cell is a bag of hemoglobin at a concentration of roughly thirty-odd grams per deciliter of cell contents, together with the enzymes that keep the iron reduced and a bath of small molecules that tune the protein's behavior. The most important tuner in humans is BPG. It binds in the central cavity of the deoxygenated hemoglobin tetramer, wedging between the two beta chains, and stabilizes the low-affinity form. The effect is to make hemoglobin release oxygen more readily. Without BPG the protein clings.

The numbers make the point. Hemoglobin inside a normal human red cell half-saturates with oxygen at a partial pressure, called the P50, of around 26 to 27 millimeters of mercury. Take the same protein out of the cell, put it in plasma, and the BPG washes away. The P50 falls to something like 12 to 14 millimeters of mercury. That is a hemoglobin that will load beautifully in the lung and then refuse to give its oxygen up until the surrounding tissue is nearly anoxic. For a carrier meant to feed starving tissue, this is a fatal design flaw.

There is a second problem. Outside the cell, the four-chain tetramer can split into two-chain halves. These dimers are small enough to slip through the kidney filter and out of the circulation, which shortens the useful life of the protein and delivers a burden of free hemoglobin to a delicate organ. So a successful carrier has to solve two problems at once: fix the oxygen affinity, and glue the pieces together.

The early workarounds

The first generation of engineers attacked the two problems by brute chemical force applied to human hemoglobin.

One approach, pyridoxylation, attaches a small phosphate-bearing group to mimic the effect of BPG and lower the affinity. Another, the diaspirin cross-link used in the product known as HemAssist, bridges the two alpha chains with a chemical clamp, locking the tetramer together and, as a side effect, shifting the P50 up toward the thirties. A third, polymerization with

glutaraldehyde, stitches many tetramers into larger assemblies, addressing the filtration problem, though it can also trap the protein in a fixed conformation.

These interventions worked, more or less, in the sense that the resulting molecules could hold together and give up oxygen. They also carried consequences. The chemical modifications were sometimes crude, and the final product was often a mixture. The trial history of the class, which I have discussed in other posts, includes serious safety signals, and it would be an oversimplification to blame any one design decision. But it is fair to say that starting from human hemoglobin meant starting from a protein whose native regulation had been taken away, and rebuilding it by hand.

The cow has a different arrangement

Now look at the ruminants. In cattle red cells, the concentration of BPG is very low, far lower than in humans. Bovine hemoglobin has evolved to work in a red cell that does not depend on BPG to tune its affinity. Instead its oxygen affinity is regulated largely by chloride ions, which are abundant in plasma at a concentration of about a hundred millimolar, and by the ordinary effects of pH and carbon dioxide.

This is a startling fact when you first take it in. A protein that is tuned by chloride is a protein that is already tuned when you put it into plasma. Chloride does not wash away. It is the most common anion in extracellular fluid. Bovine hemoglobin dissolved in a chloride-containing solution therefore has a P50 in the neighborhood of thirty or forty millimeters of mercury, which is in a physiologically useful range, without any chemical addition.

Reviewers of the field have noted this advantage repeatedly: chloride rather than BPG regulates affinity, giving bovine hemoglobin an edge over the human protein in cell-free form. It is not the only reason the bovine protein was chosen. Cattle are plentiful, slaughterhouses offer a large and regulated supply of blood, and bovine hemoglobin has been reported to be more thermally stable than the human protein. But the allosteric difference is the most elegant one, because it means the protein arrives at the manufacturer already behaving like a decent oxygen transporter.

Gluing the halves together

The chloride trick does not solve the dissociation problem. The bovine tetramer still splits into dimers in dilute solution, and dimers still leak through the kidney. The manufacturers of the two best-known bovine products solved it with glutaraldehyde, a small molecule with an aldehyde at each end that reacts with the amino groups of lysine residues on the protein surface.

Treated correctly, glutaraldehyde does two things. It forms bridges within a tetramer, holding it together, and it forms bridges between tetramers, building polymers. The resulting mixture contains molecules of many sizes. Both bovine products are named, in their formal nonproprietary names, for their average molecular mass: the product marketed for humans in some countries is hemoglobin glutamer-250 (bovine), and the veterinary product is hemoglobin glutamer-200 (bovine). The numbers refer, roughly, to average masses in kilodaltons. A single human tetramer is around 64 kilodaltons, so these products average three to four times heavier.

I emphasize the word average because it hides the interesting engineering question. A polymerization reaction is statistical. It does not produce a single molecular species. It produces a distribution, from unpolymerized tetramer up through dimers of tetramers, trimers and larger assemblies. The shape of that distribution matters enormously, and it is the reason process chemists in this field are obsessed with size-exclusion chromatography, the technique that sorts molecules by size and lets them read the distribution like a bar chart.

Why the small end of the distribution matters

The reason is the two ends of the size curve. Molecules that are too small leak through the kidney and slip out of the circulation into the space beneath the endothelial lining of vessels, where free hemoglobin scavenges nitric oxide and provokes vessel tone changes. That is one mechanism proposed for the blood pressure elevation seen with earlier products. Molecules that are too large increase the viscosity of plasma and may behave differently in the microcirculation.

So a manufacturer wants to minimize the low end, and the specifications reflect that. The process aims to reduce unpolymerized tetramer and small oligomers to a low percentage through purification steps after the reaction, commonly ultrafiltration or chromatographic removal of the smallest species. I want to be careful here: the exact release specifications for the two bovine products are proprietary in part, and figures quoted in the literature vary. What can be said safely is that a lower fraction of low-molecular-weight material was a deliberate design goal, that the products were characterized by size-exclusion analysis, and that the distribution is a critical quality attribute.

A second critical attribute is the fraction of hemoglobin in the oxidized form, methemoglobin. Ferric hemoglobin cannot bind oxygen, and it also releases heme, which is toxic. A manufacturer measures it in every lot and sets a ceiling, because it rises as the product ages. The stability of the ferrous form during storage is a reason packaging and storage conditions are chosen with such care. The engineering of shelf life turns out to be an engineering of iron chemistry.

The tradeoff hiding in the P50

Now for a controversy that shows how much remains unsettled. Bovine glutamer products have a P50 in the thirties, roughly like or slightly above that of a person's red cells. Is that ideal?

A school of thought associated with the microvascular hypothesis argues that it is not. The argument runs as follows. In a healthy circulation, oxygen leaves red cells while they are still in the arterioles, the small upstream vessels, and the arteriolar wall senses the oxygen level and constricts to protect downstream tissue from excess. A carrier with a low affinity, like a bovine glutamer, releases oxygen early, triggering that constriction reflex more strongly and reducing flow. A carrier with a high affinity, which holds oxygen until the last moment, would deliver it to the capillary bed with less premature loss, avoiding vasoconstriction and preserving the perfusion of small vessels. Under that logic, the best oxygen carrier for shock is one with a P50 well below that of blood.

The opposing view points out that a high-affinity carrier delivers oxygen only in tissue that is severely hypoxic, which is exactly where you need it in shock but is a poor design for

resuscitating a patient who is anemic and otherwise stable. A carrier with a P50 in the thirties will offload in a broader range of conditions. The disagreement is not resolved. What is worth noticing is that the P50 of a bovine product is not an accident of chemistry alone; it is a physiological bet, and the bet can be argued either way.

For a foundation thinking about which indications to advocate for, this matters. A carrier with a mid-range P50 sits comfortably in an anemia niche, such as a patient who cannot receive blood. A carrier with a very high affinity might sit better in an ischemic niche, where the task is to reach the last few oxygen-starved cells. They are different jobs, and they may call for different molecules.

The Bohr effect and the CO₂ handshake

One more piece of regulation deserves mention, because it is a beautiful piece of physiology that a cell-free carrier keeps. Hemoglobin releases oxygen more readily when the surrounding tissue is acidic and rich in carbon dioxide. That is the Bohr effect. It is why exercising muscle, which makes acid and carbon dioxide, gets extra oxygen delivered exactly where it is needed.

The Bohr effect does not depend on BPG. It depends on the protonation of particular amino acid side chains and on carbamate formation at the ends of the chains. A cell-free bovine polymer retains it, at least in part, though polymerization can dampen the response by restricting the protein's movement. This means that even outside a cell, the molecule can respond to local acidity by releasing more oxygen, which is the outcome you want in a tissue that is failing. The extent of the retained response is a measurable property, and it varies among products and among lots.

I mention it because it undermines a caricature. People sometimes describe a BHOC as a dumb sack of hemoglobin that can only hold and release oxygen according to one fixed curve. The truth is that it inherits much of the responsiveness of its parent protein, dampened by chemistry. That is a reason to take the design seriously, and a reason to measure it carefully.

Reading an oxygen curve like a map

Physiologists draw the relation between oxygen pressure and hemoglobin saturation as an S-shaped curve, and the P50 is simply the pressure at the curve's midpoint. It is more useful to think of the curve as a map with three regions.

The upper flat region, above about sixty millimeters of mercury, is the lung. Here the curve barely rises with pressure, so hemoglobin loads to near-complete saturation even if the inspired air is thin. This flatness is a safety margin: a person on a mountain at reduced pressure still loads almost fully.

The steep middle region, between roughly twenty and sixty, is the working zone of the tissues. A small drop in pressure produces a large release of oxygen. A carrier's usefulness depends on where its steep region lies relative to the pressure of the tissue it feeds. Move the curve to the left, as high-affinity molecules do, and the steep zone slides toward the lower pressures, so the molecule gives up its cargo only in the most oxygen-poor territory. Move it to the right and it gives up cargo earlier.

The lower region, below about twenty, is the reserve. A carrier with a curve that is too far to the left leaves much of its oxygen locked in this region, and no amount of extra hemoglobin will help a tissue sitting there. This is the arithmetic behind the early failure of unmodified human hemoglobin in plasma. It was not that the molecule carried too little oxygen, but that it declined to release it.

Seen this way, the choice between a bovine glutamer and a high-affinity engineered molecule is a choice about which zone to serve. And chloride, in the bovine case, simply put the curve in the middle of the working zone by default.

Questions worth putting to a developer

For readers who advocate for access, the chemistry converts into a short list of questions that separate a serious program from a hopeful one.

What is the P50 of the final product, measured in plasma-like conditions at body temperature, and how much does it vary from lot to lot?

What does the size-exclusion profile look like, and what fraction of the material sits at the small end?

What is the methemoglobin content at release and at the end of the labeled shelf life?

How much of the Bohr response survives polymerization?

None of these needs a doctorate to ask, and every serious manufacturer has the numbers.

What the accident does not fix

It would be a mistake to leave the impression that the chloride story solves the problem. It solves one problem, the affinity problem, and only partially.

It does not solve nitric oxide scavenging, which depends on the availability of free hemoglobin to the vessel wall and on the size of the molecule. It does not solve auto-oxidation, the tendency of the iron to lose its electron and become useless. It does not solve immunogenicity, since bovine hemoglobin remains a foreign protein. And it does not solve the tension between a longer circulating half-life, which favors bigger polymers, and viscosity and vascular effects, which disfavor them.

What it does is remove a whole category of chemical modification from the manufacturing process. A process that does not have to attach anything to mimic BPG has fewer reactions, fewer side products and fewer opportunities for variability. This is an unglamorous benefit, and in pharmaceutical manufacturing the unglamorous benefits are frequently the ones that decide whether a product survives scale-up.

Temperature, a footnote that becomes a chapter

Every P50 quoted in a brochure is measured at body temperature, and every P50 changes when the temperature does. Cooling shifts the curve to the left: hemoglobin holds oxygen more tightly in the cold. For a patient in shock or a surgical patient cooled on purpose, this is a modest

correction. For a donor organ resting in ice, it is a large one, and it means that a molecule tuned for warm blood behaves like a different molecule at four degrees. That is the subject of a later post, but the seed is here. The chloride trick sets the default. It does not make the curve immune to the thermometer.

A note on why nobody chose the pig

Someone will ask why cattle, not pigs or horses. I do not have a definitive published answer, and I would be suspicious of anyone who claims one. The likely reasons are practical. Cattle produce a large volume of blood per animal, there is an established slaughter and inspection infrastructure, and the biochemistry of the bovine protein, with its chloride regulation, is convenient. Porcine hemoglobin does not offer the same allosteric advantage, as far as I know, and it introduces cultural and religious considerations of its own, discussed in another post on this site. The choice of species was probably made by a combination of chemistry and logistics, and both pointed toward the cow.

What to remember

If you take one idea from this essay, take this. The success of the bovine approach was never a triumph of design so much as a gift of evolution accepted with gratitude. A cow's red cell does not use BPG, so a cow's hemoglobin is tuned by something that plasma already contains. Human hemoglobin needed to be repaired before it could be used outside its cell. Bovine hemoglobin needed only to be held together.

That distinction should shape how patient communities talk about the technology. The choice of a bovine source is not a curiosity of the marketplace. It has a chemical reason, and the reason is one a non-specialist can understand: the cow's protein works in plasma because it never needed the molecule that plasma lacks.

The rest of the story, the questions of safety, of vessel tone, of oxidation and of immune response, remains open. The chloride, at least, is settled.

Further reading

Reviews of hemoglobin-based oxygen carriers describe chloride regulation of bovine hemoglobin affinity, glutaraldehyde polymerization and the naming of the glutamer products; one accessible open-access review is titled Hemoglobin-Based Oxygen Carriers: Current State-of-the-Art and Novel Molecules. Numeric values for P50 and molecular weight distributions vary across sources and lots and should be verified against product documentation before use.